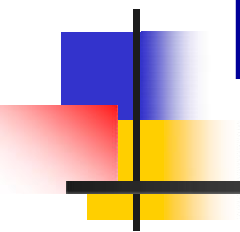




ICE 229: Discrete Mathematics

Dept. Information & Communication Engineering
Daffodil International University



A decorative graphic on the left side of the slide, consisting of a black crosshair overlaid on a blue square, a red square, and a yellow square.

Lecture 1

Course Overview

Chapter 1. The Foundations

1.1 Propositional Logic

1.1 Propositional Logic

- Logic
 - Study of reasoning.
 - Specifically concerned with whether reasoning is correct.
 - Focuses on the relationship among statements, not on the content of any particular statement.
 - Gives precise meaning to mathematical statements.
- ***Propositional Logic*** is the logic that deals with statements (propositions) and compound statements built from simpler statements using so-called *Boolean connectives*.
- Some applications in computer science:
 - Design of digital electronic circuits.
 - Expressing conditions in programs.
 - Queries to databases & search engines.

Definition of a *Proposition*

Definition: A *proposition* (denoted $p, q, r, \dots$) is simply:

- a *statement* (i.e., a declarative sentence)
 - with some definite meaning,
(not vague or ambiguous)
- having a *truth value* that's either *true* (**T**) or *false* (**F**)
 - it is **never** both, neither, or somewhere “in between!”
 - However, you might not *know* the actual truth value,
 - and, the truth value might *depend* on the situation or context.
- Later, we will study *probability theory*, in which we assign *degrees of certainty* (“between” **T** and **F**) to propositions.
 - But for now: think True/False only! (or in terms of **1** and **0**)

Examples of Propositions

- It is raining. (In a given situation)
- Beijing is the capital of China. (T)
- $2 + 2 = 5$. (F)
- $1 + 2 = 3$. (T)
- A fact-based declaration is a proposition, even if no one knows whether it is true
 - 11213 is prime.
 - There exists an odd perfect number.

Examples of Non-Propositions

The following are **NOT** propositions:

- Who's there? (interrogative, question)
- Just do it! (imperative, command)
- La la la la la. (meaningless interjection)
- Yeah, I sorta dunno, whatever... (vague)
- $1 + 2$ (expression with a non-true/false value)
- $x + 2 = 5$ (declaration about semantic tokens of non-constant value)

Truth Tables

- An *operator* or *connective* combines one or more *operand* expressions into a larger expression. (e.g., “+” in numeric expressions.)
- **Unary** operators take *one* operand (e.g., -3); **Binary** operators take *two* operands (e.g. 3×4).
- **Propositional** or **Boolean operators** operate on propositions (or their truth values) instead of on numbers.
- The **Boolean domain** is the set $\{T, F\}$. Either of its elements is called a **Boolean value**.
An n -tuple $(p_1, \dots, p_n)$ of Boolean values is called a **Boolean n -tuple**.
- An n -operand truth table is a table that assigns a Boolean value to the set of all Boolean n -tuples.

Some Popular Boolean Operators

<u>Formal Name</u>	<u>Nickname</u>	<u>Arity</u>	<u>Symbol</u>
Negation operator	NOT	Unary	$\neg$
Conjunction operator	AND	Binary	$\wedge$
Disjunction operator	OR	Binary	$\vee$
Exclusive-OR operator	XOR	Binary	$\oplus$
Implication operator	IMPLIES	Binary	$\rightarrow$
Biconditional operator	IFF	Binary	$\leftrightarrow$

The Negation Operator

- The unary ***negation operator*** “ $\neg$ ” (***NOT***) transforms a proposition into its logical *negation*.
- *E.g.* If p = “I have brown hair.”
then $\neg p$ = “It is not the case that I have brown hair” or “I do **not** have brown hair.”
- The *truth table* for NOT:

p	$\neg p$
T	F
F	T

Operand
column

Result
column

The Conjunction Operator

- The binary ***conjunction operator*** “ $\wedge$ ” (***AND***) combines two propositions to form their logical *conjunction*.

- *E.g.* If p = “I will have salad for lunch.” and q = “I will have steak for dinner.”

then, $p \wedge q$ = “I will have salad for lunch **and** I will have steak for dinner.”

Conjunction Truth Table

Operand columns

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- Note that a conjunction $p_1 \wedge p_2 \wedge \dots \wedge p_n$ of n propositions will have 2^n rows in its truth table

The Disjunction Operator

- The binary ***disjunction operator*** “ $\vee$ ” (*OR*) combines two propositions to form their logical *disjunction*.
- *E.g.* If p = “My car has a bad engine.” and q = “My car has a bad carburetor.”
then, $p \vee q$ = “My car has a bad engine, **or** my car has a bad carburetor.”

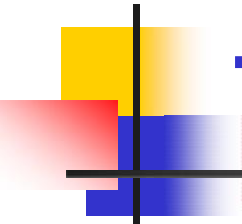
Meaning is like “and/or” in informal English.

Disjunction Truth Table

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Note difference
from AND

- Note that $p \vee q$ means that p is true, or q is true, **or both** are true!
- So, this operation is also called ***inclusive or***, because it **includes** the possibility that both p and q are true.

A decorative graphic in the top left corner consisting of overlapping yellow, red, and blue squares with a black crosshair.

The Exclusive-Or Operator

- The binary ***exclusive-or operator*** “ $\oplus$ ” (*XOR*) combines two propositions to form their logical “exclusive or”
- *E.g.* If p = “I will earn an A in this course.” and q = “I will drop this course.”, then
$$p \oplus q = \text{“I will **either** earn an A in this course, **or** I will drop it (**but not both!**)”}$$

Exclusive-Or Truth Table

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Note difference
from OR.

- Note that $p \oplus q$ means that p is true, or q is true, but **not both**!
- This operation is called ***exclusive or***, because it **excludes** the possibility that both p and q are true.

Natural Language is Ambiguous

- Note that the English “or” can be ambiguous regarding the “both” case!

- “Pat is a singer or Pat is a writer.” - ✓
- “Pat is a man or Pat is a woman.” - ⊕

p	q	p "or" q
T	T	?
T	F	T
F	T	T
F	F	F

- Need context to disambiguate the meaning!
- For this class, assume “or” means inclusive (✓).

The Implication Operator

- The conditional statement (aka *implication*) $p \rightarrow q$ states that p implies q .
- *I.e.*, If p is true, then q is true; but if p is not true, then q could be either true or false.
- *E.g.*, let p = “You study hard.”
 q = “You will get a good grade.”
 $p \rightarrow q$ = “If you study hard, then you will get a good grade.” (else, it could go either way)
 - p : *hypothesis* or *antecedent* or *premise*
 - q : *conclusion* or *consequence*

Implication Truth Table

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

The only
False case!

- $p \rightarrow q$ is **false** only when p is true but q is **not** true.
- $p \rightarrow q$ does **not** require that p or q are ever true!
- E.g. “ $(1=0) \rightarrow$ pigs can fly” is TRUE!